

Name: _____ Date: _____ Class: _____

MyDesign Portfolio

Element D Initial Template

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Name: _____ Date: _____ Class: _____

Element D: Design Concept Generation, Analysis, and Selection

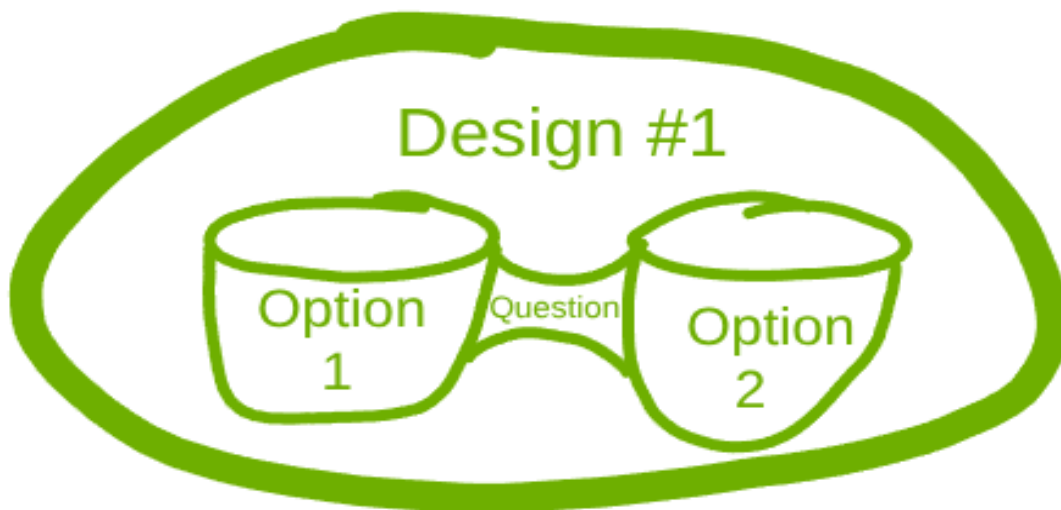
Design Idea Generation (D1)

Design Idea #1

Date: March 8th.

Description: Two simple jars held together with duct tape and glue, with an erasable whiteboard to write the question and options.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

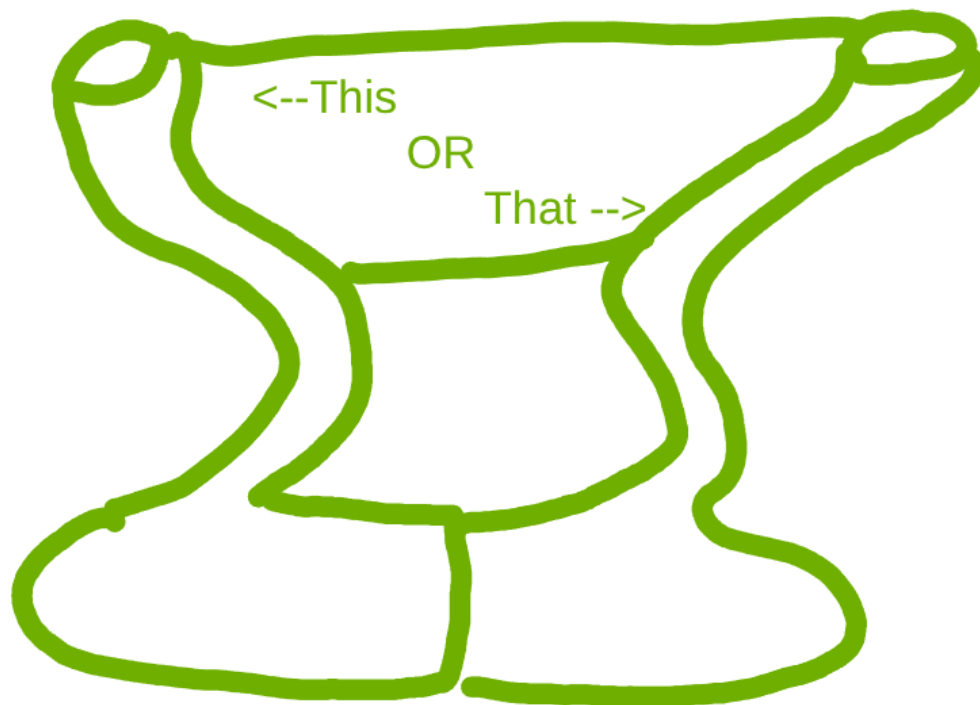


Design Idea #2

Date: March 8th

Description: A more complex design variation of Design #1, featuring see-through tubes that allow the students to see their pom-poms travel through the machine.

Drawing that includes distinguishing information such as scale, dimensions, and



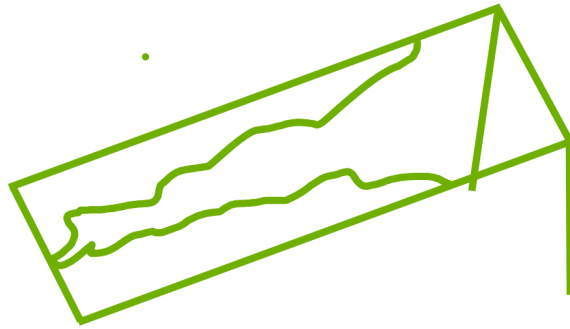
materials:

Design Idea #3

Date: March 8th

Description: This device is meant to look like a pinball machine, and has a fun twist - whoever is using it can directly see the pom-poms bonking around until they reach their destination.

Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea Comparison and Selection (D1, D2)

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Name: _____ Date: _____ Class: _____

	Option 1	Option 2	Option 3
Weight	4/5	3/5	3/5
Cost	5/5	2/5	3/5
Simplicity	5/5	2/5	4/5
Stability	5/5	4/5	4/5
score	19/20	11/20	14/20

Design Blueprint (D4)

Design Blueprint #1

Date:

Sketch:

Manufacturing Methods:

Step 1:

Step 2:

Step 3:

Supply List and Costs:

Item	Cost each	# needed	Total
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Plastic Jars	85 cents	2 needed	1.70
Plastic tubes	20 dollars	2 needed	40
Tape	6.84	1 needed	6.84

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Name: _____ **Date:** _____ **Class:** _____

Total Cost	The total cost for the materials we need is 48.54		
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Plan of Action (D3)

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

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I was unable to get the gantt template to work here but here are the primary steps in order.

1. Brainstorm how to put the machine together.
2. Make sure the process is efficient and do not waste materials when putting it together.
3. Once the machine has been put together make sure it is durable.
4. Test the machine

Here is a detailed plan of our final design, with measurements